John Vandermeulen

jackv2069@gmail.com | (803) 724-0537 | US Citizen

Education

Clemson University Clemson, SC

B.S. Mechanical Engineering, Minor in Mathematical Sciences Dec 2025 | GPA: 3.48

M.S. Mechanical Engineering (Thermal & Fluid Sciences) Dec 2026 | GPA: 3.80

Work Experience

SpaceX Bastrop, TX

Build Reliability Graduate Engineer Aug 2026-Dec 2026

- Selected to support build reliability engineering for Starlink user terminal hardware.
- Expected to collaborate with design, manufacturing, and production teams to investigate hardware issues and support validation.

Gulfstream Aerospace Corporation Savannah, GA

Cabin Systems R&D Co-op May 2026-Aug 2026

- Designed PCB schematics and layouts in Altium Designer to support fabrication of aircraft electronic hardware.
- Performed PCB assembly, soldering, electrical troubleshooting, and hardware validation.

Vendor Project Engineering Co-op May 2025-Aug 2025

- Authored 28 engineering reports supporting vendor airworthiness certification.
- Developed DO-160 qualification test plans and CATIA test fixtures for aircraft components.

Stress Engineer Co-op Aug 2024-Dec 2024

- Verified FAA Part 25 compliance of 14 G700 interior components through FEMAP analysis and hand calculations.
- Generated stress analyses and certification reports supported by Excel VBA automation.

Baseline Design Engineer Co-op Jan 2024-Apr 2024

- Designed 37 aircraft interior modifications using CATIA and ENOVIA SmarTeam.
- Worked on long term project to design a new standard interior asset for the G700 program using Catia, Kaizen walks, and weekly presentations with management.

Skills

- CAD & Design: CATIA, SolidWorks (CSWA), Onshape, AutoCAD, Altium
- Analysis: FEMAP, ANSYS Fluent, MATLAB, XFOIL, Computational Fluid Dynamics, VBA
- Software and PLM: Excel, Word, PowerPoint, SharePoint, ENOVIA SmarTeam
- Training: GD&T Training, Lean Six Sigma Yellow Belt, Advanced CATIA Training (40 hrs)

Specialized Projects

- 2D Viscous CFD Nozzle Solver: Compressible Flow, MacCormack Scheme, Body-Fitted Grid
- Composite Layup Optimization: Laminate Theory, Tsai-Wu Failure, MATLAB Automation
- Turbojet Engine Optimization: Cycle Analysis, Compressor/Turbine Matching
- Aircraft Design Project: Aerodynamics, Performance, and Stability Sizing
- Clemson Rocketry - Propulsion Sub Team: Hybrid Rocket Engine Design and Simulation